

Assignment: 10

Student Queue Management System Using Array-Based Circular Queue

Objective

The objective of this assignment is to apply the concepts of the **Queue data structure** and **Object-Oriented Programming** by developing a Student Queue Management System.

You will implement a **circular queue using an array**, where each element of the queue will be a Student object rather than a primitive data type.

The `Student` class must contain the following five attributes:

Attribute	Suggested Data Type	Description
<code>studentId</code>	<code>int</code>	Unique identification number of the student
<code>studentName</code>	<code>String</code>	Full name of the student
<code>studentAge</code>	<code>int</code>	Age of the student
<code>studentCGPA</code>	<code>double</code>	Current CGPA of the student
<code>studentInterPercentage</code>	<code>double</code>	Percentage obtained in Intermediate

Required Queue Methods Your `Queue` class must contain at least the following methods:

```
Queue(int capacity)
enqueue(Student student)
dequeue()
peek()
isEmpty()
isFull()
size()
capacity()
display()
```

The assignment will help you understand how custom objects containing multiple attributes can be stored and managed within a data structure while maintaining the **FIFO (First-In, First-Out)** principle.